NAGATA PASSES DOWN ALONG A SEPARABLE EXTENSION OF DISCRETE VALUATION RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The N-2 criterion: a discrete valuation ring is Nagata exactly when it is N-2. invoked at proof lines 35674-35677. Reduces Nagata-ness of A to finiteness of integral closures in purely inseparable extensions.
- The tensor product of a separable and a purely inseparable extension is a field. used at proof lines 35678-35681. Identifies the field in which the auxiliary integral closure B_1 is taken.
- Faithful flatness of A -> B and descent of finiteness of modules. used at proof lines 35686-35687. Carries finiteness from B tensor_A A_1 over B down to A_1 over A.

Source reference. Original lemma 03C4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4371–4371. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 032N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4026–4026. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0BRD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14766–14766. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 030U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3961–3961. Cross-chapter navigation pointer into

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Lemma 0.2. *Let $A \subset B$ be an extension of discrete valuation rings. If B is Nagata and the extension L/K of fraction fields is separable, then A is Nagata.*

Proof. A discrete valuation ring is Nagata if and only if it is N-2. Let K_1/K be a finite purely inseparable field extension. We have to show that the integral closure A_1 of A in K_1 is finite over A, see Algebra, Lemma 032N. Since L/K is separable and K_1/K is purely inseparable, the algebra $L \otimes_K K_1$ is a field (by Algebra, Lemmas 030U and 0BRD). Let B_1 be the integral closure of B in $L \otimes_K K_1$. Since B is Nagata, B_1 is finite over B. Since $B \otimes_A A_1 \subset B_1$ and B is Noetherian, we see that $B \otimes_A A_1$ is finite over B. As $A \to B$ is faithfully flat, this implies A_1 is finite over A, see Algebra, Lemma 03C4. $\square$